

IMAT MOCK EXAM 3 — "Genetics & Molecular Rotation"

60 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · personal target: 45/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "A gene is a region of DNA that encodes a functional product, usually a protein. In eukaryotes, coding regions (exons) are interrupted by non-coding regions (introns). After transcription, introns are removed and exons are spliced together to form mature messenger RNA, which is then translated by ribosomes."

Which statement is not supported?

- A) Exons are coding regions.
- B) Ribosomes transcribe DNA into RNA.
- C) Introns are removed before translation.
- D) A gene typically encodes a protein.
- E) Splicing produces mature mRNA.

Answer: B

The passage states that splicing produces mature mRNA which is then translated by ribosomes, so the only role given to ribosomes is translation. Transcription (DNA to RNA) is performed by RNA polymerase, not ribosomes, so the claim that ribosomes transcribe DNA into RNA is contradicted by the passage. Every other statement is directly supported: exons are called coding regions, introns are removed before translation, splicing forms mature mRNA, and a gene is said to usually encode a protein.

2. "A mutation is a heritable change in the nucleotide sequence of a genome. Mutations may be neutral, harmful, or — rarely — beneficial. Beneficial mutations can increase in frequency in a population over generations through natural selection, provided they confer a reproductive advantage."

The passage implies:

- A) All mutations are harmful.
- B) Neutral mutations are eliminated quickly.
- C) Mutation rate determines fitness directly.
- D) Natural selection acts on existing variation, including beneficial mutations.
- E) Beneficial mutations always become fixed.

Answer: D

The passage says beneficial mutations can increase in frequency over generations through natural selection, which only makes sense if selection acts on variation that already exists in the population. The other claims all contradict or overstate the text: it explicitly allows neutral and beneficial mutations (so not all are harmful), says only that beneficial mutations can increase in frequency (not that they always become fixed), gives no reason for neutral mutations to be quickly eliminated, and links fitness advantage to the mutation's effect rather than to the mutation rate itself.

3. (General knowledge) Which Italian scientist won the 1986 Nobel Prize in Physiology or Medicine for the discovery of nerve growth factor (NGF)?

- A) Camillo Golgi
- B) Carlo Rubbia
- C) Rita Levi-Montalcini
- D) Salvador Luria
- E) Renato Dulbecco

Answer: C

The 1986 Nobel Prize in Physiology or Medicine for the discovery of nerve growth factor (NGF) went to Rita Levi-Montalcini, shared with the American biochemist Stanley Cohen. The other Italians won in different fields and years: Golgi for his staining method revealing nerve cell structure (1906), Luria for phage genetics (1969), Dulbecco for tumour viruses (1975), and Rubbia for physics (1984), none connected to NGF.

4. In a race, Ali finished before Bashir but after Chen. Dev finished last. Who finished first?

- A) Chen
- B) Ali
- C) Dev
- D) Cannot be determined
- E) Bashir

Answer: A

Given: Ali finished before Bashir but after Chen; Dev finished last.

From Ali before Bashir: Ali is ahead of Bashir.

From Ali after Chen: Chen is ahead of Ali.

Chain so far: Chen, then Ali, then Bashir, filling places 1 to 3.

Dev is stated last, which is exactly place 4, so the order is Chen, Ali, Bashir, Dev.

Conclusion: Chen finished first.

The claim that the winner cannot be determined fails because the comparisons chain end to end and pin the entire order, leaving no arrangement in which anyone else leads.

5. "All enzymes are proteins." This statement is:

- A) True only for digestive enzymes
- B) Always true
- C) True only in prokaryotes
- D) Always false
- E) Mostly true, but some RNA molecules (ribozymes) also catalyse reactions

Answer: E

Given: the claim is that all enzymes are proteins, i.e. every member of the set of enzymes is a protein. A universal claim needs every enzyme to be a protein; a single enzyme that is not a protein makes it false.

Ribozymes are RNA molecules that catalyse reactions, and peptidyl transferase in the ribosome is RNA-based, so non-protein enzymes exist.

Therefore the claim is not strictly true, yet almost all known enzymes are proteins, so it is mostly true with known exceptions.

Conclusion: mostly true, with catalytic RNA as the exceptions.

Calling it always true overlooks ribozymes, while calling it always false ignores that the overwhelming majority of enzymes are proteins.

6. In a certain language, "TREE" is written as "USFF". How is "BIRD" written?

- A) CJTE B) AJSE C) CJSF D) BKSE E) CJSE

Answer: E

Given: TREE is written USFF; find the encoding rule and apply it to BIRD.

Compare letter by letter: T becomes U, R becomes S, E becomes F, E becomes F, so each letter moves exactly one step forward in the alphabet.

Apply to BIRD one letter at a time: B steps to C, I steps to J, R steps to S, D steps to E.

Assembling in order gives C, J, S, E.

Result: CJSE.

Any candidate differing in a single letter breaks a shift: a final F would need D to jump two steps, and a K start would need B to move backwards.

7. What comes next: 3, 6, 11, 18, 27, ?

- A) 36 B) 38 C) 39 D) 40 E) 41

Answer: B

Given: the sequence 3, 6, 11, 18, 27.

First differences: $6 - 3 = 3$, then $11 - 6 = 5$, then $18 - 11 = 7$, then $27 - 18 = 9$.

The differences are the consecutive odd numbers, so the next difference is 11.

Next term: $27 + 11 = 38$

Result: 38.

The tempting 36 repeats the last difference of 9 instead of continuing the odd-number pattern.

8. If today is Tuesday, what day will it be 100 days from today?

- A) Monday B) Wednesday C) Tuesday D) Thursday E) Friday

Answer: D

Given: today is Tuesday; count 100 days forward.

Weekdays repeat every 7 days, so only the remainder of $100 \div 7$ matters.

$100 = 14 \times 7 + 2$, so the remainder is 2

Fourteen full weeks bring the calendar back to Tuesday, then 2 more days are added.

Tuesday + 1 day = Wednesday, Tuesday + 2 days = Thursday.

Result: Thursday.

Counting Tuesday itself as the first of the two leftover days wrongly lands on Wednesday.

9. A clock reads 7:00. What is the angle between the hour and minute hands?

- A) 0° B) 90° C) 150° D) 180° E) 210°

Answer: C

Given: clock reading 7:00; dial of 360° spread over 12 hour marks.

Each hour mark spans $360^\circ \div 12 = 30^\circ$

At 7:00 the minute hand sits on 12 (0°) and the hour hand sits 7 marks away: $7 \times 30^\circ = 210^\circ$.

Raw separation: 210° .

Smaller angle: $360^\circ - 210^\circ = 150^\circ$

Result: 150° .

The tempting 210° quotes the reflex angle, forgetting that the angle between hands is the smaller arc.

Section B - Biology (Q10-32)

10. In DNA, the two strands run in opposite directions. This property is called:

- A) Antiparallelism
- B) Methylation
- C) Supercoiling
- D) Complementarity
- E) Semiconservative

Answer: A

The two strands of a DNA double helix run in opposite 5' to 3' directions: one strand runs 5' to 3' while its partner runs 3' to 5'. This geometric property is called antiparallelism. Complementarity instead refers to the A-T and G-C base pairing, supercoiling describes how the helix is coiled upon itself, and semiconservative describes the mode of replication.

11. DNA replication is described as "semiconservative" because:

- A) Both strands of the parent molecule are preserved together.
- B) Replication is conservative in prokaryotes only.
- C) Only the leading strand is copied.
- D) Each new molecule consists of one parental strand and one newly synthesised strand.
- E) The entire molecule is built anew.

Answer: D

In semiconservative replication each parental strand stays intact and acts as a template, so every daughter double helix contains one original strand and one newly synthesised strand. The Meselson and Stahl experiment confirmed exactly this. The conservative model, which would keep both parental strands together in one molecule and build a wholly new molecule in the other, is what the experiment ruled out.

12. The enzyme that joins Okazaki fragments by forming phosphodiester bonds is:

- A) Helicase
- B) Topoisomerase
- C) DNA ligase
- D) Primase
- E) DNA polymerase III

Answer: C

Okazaki fragments are short pieces synthesised discontinuously on the lagging strand. DNA polymerase extends them but cannot close the final gap, so DNA ligase is the enzyme that seals the nick by forming the phosphodiester bond between adjacent fragments. Helicase unwinds the helix, primase lays RNA primers and topoisomerase relieves supercoiling, none of which joins fragments.

13. During transcription in eukaryotes, the enzyme responsible is:

- A) DNA polymerase
- B) RNA polymerase
- C) Ligase
- D) Reverse transcriptase
- E) Helicase

Answer: B

Transcription is the synthesis of RNA from a DNA template, catalysed by RNA polymerase. DNA polymerase copies DNA during replication, reverse transcriptase makes DNA from an RNA template in retroviruses, and helicase and ligase modify DNA structure rather than synthesising RNA.

14. During which process are introns removed from pre-mRNA?

- A)** Splicing (post-transcriptional modification)
- B)** DNA repair
- C)** Translation
- D)** Transcription (initiation)
- E)** Replication

Answer: A

The primary transcript (pre-mRNA) still contains introns, which are removed by the spliceosome during post-transcriptional processing called splicing; the exons are then joined into the mature mRNA. Translation reads only the finished message and never removes introns, and replication and DNA repair act on DNA, not on RNA.

15. The three-nucleotide sequence on mRNA that specifies an amino acid is a:

- A)** Gene
- B)** Anticodon
- C)** Operon
- D)** Promoter
- E)** Codon

Answer: E

The genetic code is read on mRNA in triplets, and each three-nucleotide sequence specifying one amino acid (or a stop signal) is a codon. The anticodon is the complementary triplet located on tRNA, while a promoter is the DNA region where transcription starts and an operon is a cluster of genes transcribed together.

16. The anticodon is found on:

- A)** SnRNA
- B)** mRNA
- C)** DNA
- D)** tRNA
- E)** rRNA

Answer: D

tRNA carries the anticodon in its anticodon loop; this three-base sequence pairs with the complementary codon on mRNA so that the correct amino acid is delivered to the growing chain. mRNA carries the codons themselves and rRNA forms the structural and catalytic core of the ribosome.

17. The bonds linking amino acids in a protein are:

- A)** Peptide bonds
- B)** Hydrogen bonds
- C)** Phosphodiester bonds
- D)** Ester bonds
- E)** Glycosidic bonds

Answer: A

The carboxyl group of one amino acid reacts with the amino group of the next, releasing water and forming a peptide bond (an amide linkage) between them. Phosphodiester bonds instead link nucleotides in nucleic acids, glycosidic bonds link sugars, and hydrogen bonds only stabilise the folded three-dimensional shape.

18. The secondary structure of proteins (α -helix, β -sheet) is stabilised mainly by:

- A)** Peptide bonds
- B)** Hydrogen bonds between backbone atoms
- C)** Ionic bonds
- D)** Hydrophobic interactions
- E)** Disulfide bridges

Answer: B

Secondary structure consists of regular arrangements of the polypeptide backbone, and both the α -helix and the β -pleated sheet are stabilised by hydrogen bonds between backbone C=O and N-H groups. Disulfide bridges and hydrophobic interactions mainly stabilise the tertiary structure, and peptide bonds belong to the primary sequence.

19. A diploid cell with $2n = 6$ undergoes meiosis. How many chromosomes are in each resulting gamete?

- A)** 2 **B)** 3 **C)** 4 **D)** 6 **E)** 12

Answer: B

$2n = 6$ means the diploid cell carries two sets of 3 chromosome types, six in total. Meiosis separates homologous pairs, halving the number, so each gamete receives one set, $n = 3$ chromosomes. Keeping 6 would describe mitosis, and 12 would be a doubling rather than a halving.

20. In a cross between two heterozygous tall plants ($Tt \times Tt$), where T (tall) is dominant, what is the expected phenotypic ratio in the offspring?

- A)** 9:3:3:1
- B)** All tall
- C)** 3:1 (tall : short)
- D)** 1:2:1
- E)** 1:1

Answer: C

Step 1: the cross $Tt \times Tt$ produces genotypes TT, Tt, Tt, tt in a 1:2:1 ratio. Step 2: since T is dominant, TT and both Tt combinations are tall, so 3 of the 4 outcomes are tall and 1 is short, giving a phenotypic ratio of 3 tall : 1 short. The 1:2:1 figure is the genotypic ratio and does not describe phenotypes, while 9:3:3:1 belongs to a dihybrid cross.

21. In humans, haemophilia is an X-linked recessive trait. A carrier mother ($X^H X^h$) and an unaffected father ($X^H Y$) have a son. What is the probability the son has haemophilia?

- A)** 0 **B)** 25% **C)** 50% **D)** 75% **E)** 100%

Answer: C

A son receives the Y chromosome from his father, so his single X always comes from his mother. The carrier mother is $X^H X^h$, so each son has a 50% chance of receiving the X^h chromosome and being affected. The 25% figure would answer a different question, namely the chance per pregnancy of an affected child of unspecified sex.

22. A man who is colour-blind (X-linked recessive) has daughters with a woman who is not colour-blind and is not a carrier. What is true of their daughters?

- A)** None will carry the allele.
- B)** All will be colour-blind.
- C)** All will be colour-blind and carriers.
- D)** 50% colour-blind, 50% carriers.
- E)** All will be carriers (heterozygous) but not colour-blind.

Answer: E

The colour-blind father has only one X chromosome, and he passes that chromosome (bearing the recessive allele) to every daughter. The mother, who is neither colour-blind nor a carrier, always passes a normal X. Every daughter is therefore heterozygous X^cX^N : a carrier whose normal second X masks the recessive allele, so none of them is colour-blind.

23. In a dihybrid cross ($AaBb \times AaBb$) with independent assortment, the expected phenotypic ratio is:

- A)** 1:1 **B)** 1:2:1 **C)** All dominant **D)** 3:1 **E)** 9:3:3:1

Answer: E

Step 1: for one gene, $Aa \times Aa$ gives a $3/4$ dominant to $1/4$ recessive phenotypic split. Step 2: with independent assortment the two genes multiply: $(3/4)(3/4) = 9/16$ show both dominant traits, $(3/4)(1/4) = 3/16$ for each of the two single-recessive classes, and $(1/4)(1/4) = 1/16$ show both recessive traits. This yields the classic 9:3:3:1 ratio.

24. A bacterium acquires antibiotic resistance via a plasmid from another bacterium. This is an example of:

- A)** Meiosis
- B)** Mutation
- C)** Mitosis
- D)** Conjugation (horizontal gene transfer)
- E)** Transcription

Answer: D

Receiving a plasmid from another bacterium is conjugation, the classic case of horizontal gene transfer: DNA moves sideways between two living cells rather than being inherited from a parent cell. Mutation is a change within a cell's own genome, while mitosis, meiosis and transcription are cell division or gene expression processes that do not transfer DNA between cells.

25. The phases of mitosis in correct order are:

- A)** Prophase, metaphase, anaphase, telophase
- B)** Prophase, anaphase, metaphase, telophase
- C)** Telophase, anaphase, metaphase, prophase
- D)** Metaphase, prophase, anaphase, telophase
- E)** Anaphase, metaphase, prophase, telophase

Answer: A

Mitosis proceeds prophase (chromosomes condense, spindle forms), metaphase (chromosomes align on the equatorial plate), anaphase (sister chromatids separate and move to the poles), telophase (nuclear envelopes reform). Any ordering that puts anaphase before metaphase, or telophase first, contradicts these events, so the sequence is prophase, metaphase, anaphase, telophase.

26. Which organelle contains digestive enzymes and functions in intracellular digestion?

- A) Mitochondrion B) Chloroplast C) Nucleolus D) Ribosome E) Lysosome

Answer: E

Lysosomes are membrane-bound vesicles filled with acid hydrolases that break down worn-out organelles and material taken into the cell, which is intracellular digestion. Ribosomes build proteins, mitochondria and chloroplasts produce ATP or sugars, and the nucleolus assembles ribosomal subunits.

27. Which structure is common to both plant and animal cells?

- A) Plasmodesmata
B) Chloroplast
C) Mitochondrion
D) Large central vacuole
E) Cell wall

Answer: C

Mitochondria, the site of aerobic respiration, occur in both plant and animal cells. The cell wall, the large central vacuole, chloroplasts and plasmodesmata are features of plant cells and are absent from animal cells, so only the mitochondrion is shared.

28. In cellular respiration, glycolysis produces a net gain of:

- A) 0 ATP (all invested)
B) 2 ATP, 2 NADH, 2 pyruvate (per glucose)
C) 2 ATP, 2 FADH₂, 2 pyruvate
D) 4 ATP, 4 NADH, 2 pyruvate
E) 38 ATP

Answer: B

Glycolysis invests 2 ATP in its preparatory phase and regenerates 4 ATP in its payoff phase, for a net gain of 2 ATP, together with 2 NADH and 2 pyruvate molecules per glucose. The claim of 4 ATP quotes the gross production and forgets the 2 consumed, and FADH₂ is produced by the Krebs cycle, not by glycolysis.

29. The non-protein component required for some enzymes to function (e.g., a metal ion) is called a:

- A) Cofactor (or coenzyme if organic)
B) Apoenzyme
C) Competitive inhibitor
D) Substrate
E) Allosteric inhibitor

Answer: A

Many enzymes need a non-protein helper to be active: an inorganic ion such as Zn²⁺ or Mg²⁺ is called a cofactor, while an organic helper molecule is a coenzyme. The apoenzyme is the inactive protein part without its helper, and the substrate is the molecule the enzyme acts on, so neither of those fits the description.

30. At the neuromuscular junction, the neurotransmitter released by motor neurons is:

- A) GABA B) Serotonin C) Glutamate D) Acetylcholine E) Dopamine

Answer: D

Motor neurons release acetylcholine at the neuromuscular junction; it binds receptors on the muscle fibre membrane and triggers depolarisation and contraction. Dopamine, GABA, serotonin and glutamate are neurotransmitters of the central nervous system, not the junction with skeletal muscle.

31. The resting potential of a typical neuron is approximately:

- A) -70 mV B) -30 mV C) 0 mV D) $+30$ mV E) $+70$ mV

Answer: A

In a resting neuron the inside of the membrane sits about 70 mV below the outside, maintained by potassium leak channels and the Na^+/K^+ pump, so the resting potential is approximately -70 mV. The value $+30$ mV instead corresponds to the peak of an action potential, not to the resting state.

32. Which vitamin is required for blood clotting?

- A) Vitamin D B) Vitamin A C) Vitamin K D) Vitamin C E) Vitamin B12

Answer: C

Vitamin K is required for the gamma-carboxylation of clotting proteins such as prothrombin (its very name comes from Koagulation), so without it blood fails to clot properly. Vitamin C is involved in collagen synthesis, vitamin D in calcium absorption, vitamin A in vision, and vitamin B12 in red blood cell formation.

Section C - Chemistry (Q33-47)

33. How many moles are in 36 g of water (H_2O ; molar mass = 18 g/mol)?

- A) 0.5 B) 1 C) 2 D) 18 E) 36

Answer: C

Given: mass $m = 36$ g of H_2O ; molar mass $M = 18$ g/mol

Relationship: amount of substance $n = m \div M$.

Substitute: $n = 36 \text{ g} \div 18 \text{ g/mol}$

$$36 \div 18 = 2$$

Result: $n = 2$ mol

Classic slip: 0.5 mol, from dividing the molar mass by the mass ($18 \div 36$) instead of mass by molar mass.

34. Balance: $__ \text{C}_3\text{H}_8 + __ \text{O}_2 \rightarrow __ \text{CO}_2 + __ \text{H}_2\text{O}$. The coefficient of O_2 is:

- A) 3 B) 4 C) 5 D) 6 E) 8

Answer: C

Given: skeleton equation $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$.

Method: balance one element at a time, in the order C, then H, then O.

Carbon: 3 C atoms in C_3H_8 require 3 CO_2 .

Hydrogen: 8 H atoms in C_3H_8 require $8 \div 2 = 4$ H_2O

Oxygen: products carry $3 \times 2 + 4 \times 1 = 10$ O atoms, so O_2 needed = $10 \div 2 = 5$.

Result: $\text{C}_3\text{H}_8 + 5 \text{O}_2 \rightarrow 3 \text{CO}_2 + 4 \text{H}_2\text{O}$, so the O_2 coefficient is 5 .

Classic slip: 4 , from copying the water coefficient across to O_2 instead of counting oxygen atoms on both sides.

35. The oxidation number of nitrogen in NO_3^- is:

- A) -3 B) +3 C) +4 D) +5 E) +6

Answer: D

Given: nitrate ion NO_3^- carrying charge -1; standard assignment O = -2.

Relationship: oxidation numbers sum to the ion charge: $x(\text{N}) + 3 \times x(\text{O}) = -1$.

Substitute: $x + 3 \times (-2) = -1$

$3 \times (-2) = -6$

$x - 6 = -1$

$x = -1 + 6 = +5$

Result: nitrogen is +5.

Classic slip: +6, from balancing the sum to zero and forgetting that the ion carries charge -1.

36. What is the pH of a 0.001 M solution of HNO_3 (strong acid)?

- A) 1 B) 2 C) 3 D) 11 E) 13

Answer: C

Given: HNO_3 at 0.001 M; nitric acid is strong, so it dissociates completely.

Relationship: one H^+ per acid molecule gives $[\text{H}^+] = 0.001 \text{ M} = 10^{-3} \text{ M}$, and $\text{pH} = -\log_{10}[\text{H}^+]$.

Substitute: $\text{pH} = -\log_{10}(10^{-3})$

$\log_{10}(10^{-3}) = -3$, so $\text{pH} = -(-3) = 3$

Result: pH 3.

Classic slip: 11, from applying the base formula $14 - \text{pH}$ as if the solution were alkaline.

37. Which of the following is a weak acid?

- A) HNO_3
B) HBr
C) HCl
D) H_2SO_4
E) CH_3COOH (acetic acid)

Answer: E

HCl , HNO_3 , H_2SO_4 and HBr are all strong acids that dissociate essentially completely in water. Acetic acid, CH_3COOH , is the only weak acid in the list: it establishes an equilibrium in which most molecules remain undissociated.

38. Which bond is most polar?

- A) C-H B) O-H C) H-H D) C-C E) N-H

Answer: B

Bond polarity grows with the electronegativity difference between the two atoms. Using Pauling values, O (3.44) versus H (2.20) differs by about 1.2, N (3.04) versus H by about 0.8, C (2.55) versus H by about 0.4, while C-C and H-H join identical atoms and are completely non-polar. The O-H bond therefore shows the largest charge separation.

39. Boyle's law states that, at constant temperature, the pressure of a fixed mass of gas is:

- A) Independent of volume
- B) Inversely proportional to its volume
- C) Proportional to temperature
- D) Directly proportional to its volume
- E) Proportional to moles

Answer: B

Boyle's law states $pV = \text{constant}$ for a fixed amount of gas at constant temperature, which means pressure is inversely proportional to volume: doubling the volume halves the pressure. Direct proportionality to volume contradicts the law, and proportionality to temperature or moles describes different, non-isothermal or non-fixed-amount conditions.

40. The number of protons in an atom of ^{23}Na (sodium) is:

- A) 11 B) 12 C) 22 D) 23 E) 34

Answer: A

The mass number 23 counts protons plus neutrons and changes between isotopes, but the proton number is fixed by the element's atomic number. Sodium's atomic number is 11, so every sodium atom has 11 protons (and in this isotope $23 - 11 = 12$ neutrons). The answers 23 or 12 confuse the mass number or neutron count with the proton count.

41. Which of the following is an alkene?

- A) C_3H_8 B) C_6H_6 C) CH_4 D) C_2H_4 E) C_2H_6

Answer: D

Alkenes contain one $\text{C}=\text{C}$ double bond and follow the general formula C_nH_{2n} . C_2H_4 (ethene) fits this formula exactly and is the alkene. CH_4 , C_2H_6 and C_3H_8 are saturated alkanes fitting $\text{C}_n\text{H}_{2n+2}$, and C_6H_6 is the aromatic benzene ring, which is not classified as an alkene.

42. The functional group of alcohols is:

- A) $-\text{CHO}$ B) $-\text{NH}_2$ C) $-\text{COOH}$ D) $-\text{CO}-$ E) $-\text{OH}$

Answer: E

Alcohols are defined by a hydroxyl group, $-\text{OH}$, bonded to a saturated carbon, as in ethanol $\text{CH}_3\text{CH}_2\text{OH}$. The $-\text{CHO}$ group marks aldehydes, $-\text{COOH}$ carboxylic acids, $-\text{CO}-$ ketones, and $-\text{NH}_2$ amines.

43. Which has a $\text{C}=\text{O}$ group bonded to two carbons (within a chain)?

- A) Carboxylic acid
- B) Ketone
- C) Alcohol
- D) Aldehyde
- E) Amine

Answer: B

A carbonyl group ($\text{C}=\text{O}$) whose carbon is bonded to two other carbons inside the chain is the defining feature of a ketone, as in propanone $\text{CH}_3\text{-CO-CH}_3$. In an aldehyde the carbonyl carbon is terminal and carries a hydrogen, and in a carboxylic acid it is additionally bonded to an $-\text{OH}$.

44. Increasing the concentration of a reactant in a reaction generally:

- A) Decreases activation energy
- B) Decreases the rate
- C) Shifts K_{eq}
- D) Increases the rate (more frequent collisions)
- E) Has no effect on rate

Answer: D

A higher reactant concentration means more particles per unit volume, so collisions occur more frequently and more particles per second possess the energy and orientation needed to react, raising the rate. Concentration does not change the activation energy, which is a fixed property of the reaction pathway, and it does not shift the equilibrium constant K , which only temperature can change.

45. For an endothermic reaction, ΔH is:

- A) Positive
- B) Zero
- C) Equal to activation energy
- D) Negative
- E) Always equal to ΔS

Answer: A

ΔH measures the enthalpy change of the system; endothermic means the system absorbs heat, so the products end up higher in enthalpy than the reactants and ΔH is positive. The sign convention is the reverse for exothermic reactions. ΔH also cannot equal ΔS (a quantity in J/K) or the activation energy, which measures the barrier height rather than the overall energy change.

46. Which isotope is radioactive?

- A) ^{13}C B) ^{16}O C) ^{14}N D) ^{12}C E) ^{14}C

Answer: E

Carbon-14 contains two more neutrons than the stable isotopes ^{12}C and ^{13}C and decays by beta emission with a half-life of about 5730 years, which is the basis of radiocarbon dating. ^{16}O and ^{14}N are stable nuclides, so carbon-14 is the only radioactive species listed.

47. A solution has a pH of 8. It is:

- A) Strongly basic
- B) Strongly acidic
- C) Weakly acidic
- D) Neutral
- E) Weakly basic

Answer: E

At 25 °C neutrality corresponds to pH 7, where $[\text{H}^+] = 10^{-7} \text{ M}$

A pH of 8 means $[\text{H}^+] = 10^{-8} \text{ M}$, ten times fewer hydrogen ions than neutral water, so the solution is mildly (weakly) basic. Strongly basic solutions lie well above pH 10, and anything below 7 would be acidic.

Section D - Mathematics (Q48-54)

48. Simplify: $(2^3) \times (2^2)$.

- A) $2^5 (= 32)$ B) 2^6 C) $2^1 (= 2)$ D) 4^5 E) 8

Answer: A

Given: the product $2^3 \times 2^2$.

Relationship: multiplying powers of the same base adds exponents, $a^m \times a^n = a^{m+n}$.

Substitute: $2^3 \times 2^2 = 2^{3+2} = 2^5$

$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

Check directly: $2^3 = 8$ and $2^2 = 4$, and $8 \times 4 = 32$

Result: 32.

Classic slip: 2^6 , from multiplying the exponents 3×2 instead of adding them.

49. Solve: $\log_{10}(x) = 2$.

- A) $x = 20$ B) $x = 100$ C) $x = 10$ D) $x = 2$ E) $x = 1000$

Answer: B

Given: $\log_{10}(x) = 2$

Relationship: $\log_{10}(x) = y$ is equivalent to $x = 10^y$

Rewrite in exponential form: $x = 10^2$

$10^2 = 10 \times 10 = 100$

Check: $\log_{10}(100) = 2$ because $10^2 = 100$

Result: $x = 100$

Classic slip: $x = 20$, from multiplying 10 by the exponent 2 instead of raising 10 to the power 2.

50. Solve: $x^2 = 49$ ($x > 0$).

- A) 6 B) 7 C) 8 D) 14 E) 24.5

Answer: B

Given: $x^2 = 49$ with the condition $x > 0$

Relationship: $x^2 = k$ gives $x = +\sqrt{k}$ or $x = -\sqrt{k}$

$\sqrt{49} = 7$ because $7 \times 7 = 49$

The other root is -7 , but the condition $x > 0$ removes it.

Result: $x = 7$

Classic slip: 14, from doubling 7 instead of taking the square root.

51. Solve the inequality: $x/3 - 1 \geq 2$.

- A) $x \geq 6$ B) $x \geq 1$ C) $x \geq 12$ D) $x \geq 9$ E) $x \geq 3$

Answer: D

Given: $x/3 - 1 \geq 2$.

Isolate the x term: add 1 to both sides, giving $x/3 \geq 2 + 1 = 3$.

Clear the fraction: multiply both sides by 3; since $3 > 0$ the direction does not flip, giving $x \geq 3 \times 3 = 9$.

Boundary check: at $x = 9$ the left side is $9 \div 3 - 1 = 3 - 1 = 2$, which satisfies ≥ 2 exactly.

Result: $x \geq 9$.

Classic slip: $x \geq 3$, from stopping after the first step and never clearing the division by 3.

52. A square has perimeter 20 cm. What is its area?

- A) 5 cm² B) 10 cm² C) 20 cm² D) 25 cm² E) 100 cm²

Answer: D

Given: square with perimeter 20 cm.

Relationship: perimeter = $4 \times \text{side}$, so side = perimeter $\div 4$.

Substitute: side = $20 \text{ cm} \div 4 = 5 \text{ cm}$

Area relationship: area = side $\times$ side.

Substitute: area = $5 \text{ cm} \times 5 \text{ cm} = 25 \text{ cm}^2$

Result: 25 cm².

Classic slip: 100 cm², from squaring the perimeter 20 instead of the side 5.

53. $\cos 60^\circ = ?$

- A) 0 B) $\sqrt{3}/2$ C) $1/2$ D) $\sqrt{2}/2$ E) 1

Answer: C

Given: the angle 60° .

Relationship: in a 30° - 60° - 90° triangle the sides opposite those angles are in ratio $1 : \sqrt{3} : 2$, and cosine = adjacent $\div$ hypotenuse.

For 60° the adjacent side is the shortest leg, length 1, and the hypotenuse is 2.

Substitute: $\cos 60^\circ = 1 \div 2 = 1/2$

Result: $1/2$.

Classic slip: $\sqrt{3}/2$, which is $\cos 30^\circ$, from swapping which leg is adjacent to the 60° angle.

54. Two fair coins are tossed. What is the probability of getting at least one head?

- A) $1/4$ B) $1/2$ C) 1 D) $1/3$ E) $3/4$

Answer: E

Given: two independent tosses of a fair coin; event = at least one head.

Sample space: HH, HT, TH, TT, four equally likely outcomes.

Favourable outcomes: HH, HT, TH, three of the four.

Substitute: $P(\text{at least one head}) = 3 \div 4 = 3/4$

Check by complement: $P(\text{no head}) = P(\text{TT}) = 1/4$, and $1 - 1/4 = 3/4$

Result: $3/4$.

Classic slip: $1/2$, from reasoning as though only one toss needed to come up heads and the other could be ignored.

Section E - Physics (Q55-60)

55. A car decelerates uniformly from 20 m/s to rest in 5 s. What is its deceleration?

- A) 2 m/s² B) 4 m/s² C) 5 m/s² D) 10 m/s² E) 20 m/s²

Answer: B

Given: initial speed $u = 20$ m/s, final speed $v = 0$ m/s (rest), time $t = 5$ s.

Relationship: uniform acceleration $a = (v - u) \div t$.

Substitute: $a = (0 - 20 \text{ m/s}) \div 5 \text{ s}$

$$0 - 20 = -20 \text{ m/s}$$

$$-20 \div 5 = -4 \text{ m/s}^2$$

Result: the acceleration is -4 m/s², so the deceleration is 4 m/s².

Classic slip: 2 m/s², from dividing the average speed ($20 \div 2 = 10$ m/s) by the time instead of the speed change.

56. Three 6 Ω resistors are connected in parallel. The equivalent resistance is:

- A) 2 Ω B) 3 Ω C) 6 Ω D) 9 Ω E) 18 Ω

Answer: A

Given: three resistors, each $R = 6 \Omega$, connected in parallel.

Relationship: $1/R_{\text{eq}} = 1/R_1 + 1/R_2 + 1/R_3$

Substitute: $1/R_{\text{eq}} = 1/6 + 1/6 + 1/6$ (in $1/\Omega$)

$$1/6 + 1/6 + 1/6 = 3/6 = 1/2$$

$$R_{\text{eq}} = 1 \div (1/2) = 2 \Omega$$

Result: 2 Ω .

Classic slip: 18 Ω , from adding the three resistances as if in series; for equal resistors in parallel the quick check is $R \div n = 6 \div 3 = 2 \Omega$.

57. What is the voltage across a 10 Ω resistor carrying 0.5 A?

- A) 0.05 V B) 5 V C) 10 V D) 20 V E) 50 V

Answer: B

Given: resistance $R = 10 \Omega$, current $I = 0.5$ A

Relationship: Ohm's law $V = I \times R$.

Substitute: $V = 0.5 \text{ A} \times 10 \Omega$

$$0.5 \times 10 = 5$$

Result: $V = 5$ V

Classic slip: 20 V, from computing $10 \div 0.5$, dividing by the current instead of multiplying.

58. A ball of mass 0.5 kg moves at 4 m/s. What is its kinetic energy?

- A) 1 J B) 2 J C) 4 J D) 8 J E) 16 J

Answer: C

Given: mass $m = 0.5$ kg, speed $v = 4$ m/s

Relationship: kinetic energy $KE = \frac{1}{2} \times m \times v^2$

Substitute: $KE = \frac{1}{2} \times 0.5 \text{ kg} \times (4 \text{ m/s})^2$

$$(4 \text{ m/s})^2 = 16 \text{ m}^2/\text{s}^2$$

$$\frac{1}{2} \times 0.5 \text{ kg} = 0.25 \text{ kg}$$

$$KE = 0.25 \text{ kg} \times 16 \text{ m}^2/\text{s}^2 = 4 \text{ J}$$

Result: 4 J.

Classic slip: 2 J, from computing $m \times v = 0.5 \times 4$ without squaring the speed.

59. A 10 N force moves an object 5 m in the direction of the force. How much work is done?

- A) 0 J B) 2 J C) 5 J D) 15 J E) 50 J

Answer: E

Given: force $F = 10$ N, displacement $d = 5$ m in the direction of the force.

Relationship: work $W = F \times d$.

Substitute: $W = 10 \text{ N} \times 5 \text{ m}$

$$10 \times 5 = 50$$

Result: $W = 50$ J

Classic slip: 2 J, from dividing 10 by 5 instead of multiplying.

60. Specific heat capacity of water is 4200 J/(kg·°C). How much energy heats 2 kg of water by 10 °C?

- A) 8,400 J B) 42,000 J C) 84,000 J D) 420,000 J E) 840 J

Answer: C

Given: mass $m = 2$ kg, specific heat capacity $c = 4200$ J/(kg·°C), temperature rise $\Delta T = 10$ °C.

Relationship: heat energy $Q = m \times c \times \Delta T$.

Substitute: $Q = 2 \text{ kg} \times 4200 \text{ J}/(\text{kg} \cdot ^\circ\text{C}) \times 10 \text{ }^\circ\text{C}$

$$2 \times 4200 = 8400 \text{ J}/^\circ\text{C}$$

$$8400 \text{ J}/^\circ\text{C} \times 10 \text{ }^\circ\text{C} = 84,000 \text{ J}$$

Result: 84,000 J.

Classic slip: 42,000 J, from dropping the factor of 2 and computing for 1 kg of water.